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Northwestern Pacific Oceanic circulation shaped by ENSO

You-Lin Wang[1]✉, Fei-Fei Jin[2], Chau-Ron Wu[1,3]✉ & Bo Qiu[4]

The intricate currents of the Northwest Pacific Ocean, with strong manifestations along the westside rim, connect tropical and subtropical gyres and significantly influence East Asian and global climates. The El Niño/Southern Oscillation (ENSO) originates in the tropical Pacific Ocean and disrupts this ocean circulation system. However, the spatiotemporal dependence of the impact of ENSO events has yet to be elucidated because of the complexities of both ENSO events and circulation systems, as well as the increased availability of observational data. We thus combined altimeter and drifter observations to demonstrate the distinct tropical and subtropical influences of the circulation system on ENSO diversity. During El Niño years, the North Equatorial Current, North Equatorial Countercurrent, Mindanao Current, Indonesian Throughflow, and the subtropical Kuroshio Current and its Extension region exhibit strengthening, while the tropical Kuroshio Current weakens. The tropical impact is characterized by sea level changes in the warm pool, whereas the subtropical influence is driven by variations in the wind stress curl. The tropical and subtropical influences are amplified during the Centra Pacific El Niño years compared to the Eastern Pacific El Niño years. As the globe warms, these impacts are anticipated to intensify. Thus, strengthening observation systems and refining climate models are essential for understanding and projecting the enhancing influences of ENSO on the Northwest Pacific Oceanic circulation.

The Northwest Pacific Ocean features intricate currents that shape tropical and subtropical gyres (Fig. 1)[1,2]. Within the tropical gyre, the North Equatorial Current bifurcates east of Luzon, forming the tropical Kuroshio Current flowing north and the Mindanao Current flowing south[3,4]. Subsequently, the Mindanao Current merges with the North Equatorial Countercurrent, flowing east into the central Pacific or south via the Indonesian Throughflow into the Indian Ocean[5,6]. In the subtropical gyre, the Kuroshio Current travels north from eastern Taiwan through the East China Sea, reaching the Tokara Strait and forming the Kuroshio Extension in the mid-latitude central Pacific (Fig. 1)[7]. The modern understanding of wind-driven ocean circulation, rooted in Pacific currents, has advanced due to new insights into the impact of the Earth's rotation on gyres and Ekman transport. These insights emphasize the role of wind stress gradients on western boundary currents and define Pacific wind-driven circulations: Tropical and subtropical gyres[2].

The circulation patterns in the Northwest Pacific Ocean significantly impact East Asia and the global climate[2,8,9]. In the tropics, the Indonesian Throughflow links ocean basins at low latitudes, integrating water from the Mindanao Current via wind-driven circulation (Fig. 1)[6]. It serves as a vital component of thermohaline circulation and is crucial in regulating global climate dynamics[6]. Additionally, the Pacific Ocean hosts the El Niño/Southern Oscillation (ENSO), a key driver of global climate variability. ENSO disrupts weather patterns, affecting ecosystems, agriculture, cyclones, droughts, wildfires, and floods worldwide[10–12]. The Pacific tropical gyre transports masses toward the equator, greatly influencing the Western Pacific warm pool, ENSO life cycle, and Asian monsoon system[2,13]. In the subtropics, the interaction between mid-latitude cold air and warm water along the Kuroshio Extension facilitates the exchange of heat and moisture between the ocean and the atmosphere, thus influencing winter storm tracks[14]. Moreover, North Pacific coupled ocean–atmosphere dynamics regulate El Niño development: The atmospheric response to the Kuroshio Extension triggers the Pacific Meridional Modes that drive the ENSO, and ENSO teleconnections reinforce the extratropical atmospheric forcing of the Kuroshio Extension[15]. These findings have highlighted the importance of the Northwest Pacific Ocean circulation. Consequently, exploring the response of the Northwest Pacific Ocean circulation under the influence of climate change has attracted increased attention.

On the interannual timescale, extensive research findings have suggested that ENSO exerts a dominant influence on regulating Northwest Pacific Ocean circulation[2,16]. In the tropics, El Niño strengthens the North

[1]Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan. [2]Department of Atmospheric Sciences, University of Hawaii at Manoa, Hawaii, USA. [3]Department of Earth Sciences, National Taiwan Normal University, Taipei, Taiwan. [4]Department of Oceanography, University of Hawaii at Manoa, Hawaii, USA. ✉email: youlinwang@gate.sinica.edu.tw; cwu@ntnu.edu.tw